

SIMATS ENGINEERING

Name	Register Number	Course Code	Course
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14. Perform transformation using Homography matrix.

AIM:

Transformation using Homography matrix.

PROGRAM:

```
import cv2
import numpy as np

img = cv2.imread(r'C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png')

if img is None:
    print("Error: Image not found!")
else:
    rows, cols = img.shape[:2]

    src_points = np.float32([
        [0, 0],
        [cols - 1, 0],
        [0, rows - 1],
        [cols - 1, rows - 1]
    ])

    dst_points = np.float32([
        [0, 0],
        [cols - 1, 0],
        [0, int(0.7 * rows)],
        [cols - 1, int(0.7 * rows)]
    ])

    M, _ = cv2.findHomography(src_points, dst_points)

    homography_img = cv2.warpPerspective(img, M, (cols, rows))

    cv2.imwrite(
        r'C:\Users\Susmitha\Downloads\ITA0510
        Lab\transformation_using_Homography_Image.jpg',
```

```
homography_img  
)  
  
cv2.imshow("Homography Transformed Image", homography_img)  
  
cv2.waitKey(0)  
cv2.destroyAllWindows()
```

INPUT:



OUTPUT:



RESULT:

Thus the program is executed successfully.